

ERRACHIDIA MOULAY ALI CHERIF, Morocco

WMO: 602100

Lat: 31.948N

Lon: 4.398W

Elev: 1045

StdP: 89.39

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.9	(c) 2.2	(d) -15.7	(e) 1.1	(f) 21.7	(g) -12.9	(h) 1.4	(i) 21.4	(j) 11.0	(k) 10.3	(l) 9.1	(m) 11.6	(n) 0.6	(o) 20	(p) 0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.3	(c) 40.1	(d) 18.1	(e) 39.2	(f) 18.1	(g) 38.2	(h) 18.0	(i) 22.6	(j) 35.2	(k) 21.4	(l) 34.4	(m) 20.5	(n) 33.7	(o) 4.8	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	15.2	30.1	17.1	13.9	27.9	15.9	12.9	27.0	72.2	35.0	67.6	34.7	64.1	33.9	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	9.7	8.0	-1.3	41.4	1.7	0.9	-2.5	42.1	-3.5	42.6	-4.4	43.1	-5.7	43.8
			-2.8	22.9	1.9	2.3	-4.2	24.6	-5.3	26.0	-6.4	27.3	-7.8	29.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.5	9.4	11.5	15.7	19.7	24.1	28.7	32.7	31.3	26.5	20.8	14.4	10.3	(6)
(7)		DBStd	8.40	2.56	2.60	3.06	3.06	3.60	2.74	1.98	2.31	3.01	2.63	3.11	2.38	(7)
(8)		HDD10.0	85	43	13	1	0	0	0	0	0	0	0	3	26	(8)
(9)		HDD18.3	965	278	191	91	21	3	0	0	0	0	9	123	250	(9)
(10)		CDD10.0	3908	23	56	179	289	436	562	705	659	495	335	134	34	(10)
(11)		CDD18.3	1746	0	0	10	61	181	312	447	401	245	86	4	0	(11)
(12)		CDH23.3	24633	1	3	123	577	2099	4436	7457	6157	3026	711	43	1	(12)
(13)		CDH26.7	14288	0	0	14	151	951	2556	5063	3877	1516	159	2	0	(13)
(14)	Wind	WSAvg	2.7	1.5	2.1	2.9	3.5	3.8	3.5	3.6	3.5	3.0	2.2	1.7	1.4	(14)
(15)	Precipitation	PrecAvg	136	8	12	13	9	9	6	3	6	17	22	20	9	(15)
(16)		PrecMax	279	33	49	42	66	45	45	28	20	76	106	172	69	(16)
(17)		PrecMin	57	0	0	1	0	0	0	0	0	1	0	0	0	(17)
(18)		PrecStd	58	11	15	12	16	12	10	6	5	19	28	37	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.5	23.3	28.4	32.0	36.5	39.7	41.6	40.6	37.6	31.7	26.7	22.2	(19)
(20)			MCWB	10.2	10.7	12.8	13.2	14.7	17.6	16.5	18.5	17.1	14.8	13.5	10.1	(20)
(21)		2%	DB	19.4	21.6	26.4	30.1	34.9	38.2	40.7	39.4	36.1	29.8	24.6	20.0	(21)
(22)			MCWB	9.6	10.4	12.1	13.2	15.3	16.8	17.5	18.6	17.1	15.7	13.7	9.6	(22)
(23)		5%	DB	17.8	20.0	24.6	28.4	33.2	37.1	39.8	38.5	34.8	28.3	23.0	18.2	(23)
(24)			MCWB	8.7	9.7	11.5	13.0	15.4	16.8	18.0	18.7	17.2	15.4	12.7	9.8	(24)
(25)		10%	DB	16.1	18.5	23.0	26.7	31.7	35.7	38.9	37.4	33.4	27.1	21.3	16.5	(25)
(26)			MCWB	7.8	9.2	11.1	12.7	15.2	16.4	18.0	18.8	17.3	15.1	11.7	9.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	13.5	15.6	17.3	20.0	21.7	24.1	24.3	22.1	19.7	16.9	13.0	(27)
(28)			MCDWB	19.3	20.3	25.1	26.2	31.0	34.8	38.1	37.2	32.6	26.3	24.4	18.2	(28)
(29)		2%	WB	10.8	12.1	14.2	16.1	18.7	20.3	22.4	22.8	20.7	18.2	15.0	11.9	(29)
(30)			MCDWB	17.2	18.9	23.5	25.4	29.7	34.0	36.7	35.0	31.3	25.8	21.4	16.7	(30)
(31)		5%	WB	9.7	11.1	13.1	15.0	17.7	19.3	21.2	21.7	19.7	17.3	13.7	11.0	(31)
(32)			MCDWB	15.9	17.5	22.0	24.2	28.4	32.7	36.2	33.8	30.2	24.8	21.2	15.8	(32)
(33)		10%	WB	8.6	10.1	12.1	14.0	16.7	18.3	20.2	20.7	18.9	16.6	12.7	9.9	(33)
(34)			MCDWB	14.7	16.4	20.7	23.2	27.4	31.3	35.4	33.5	29.3	23.9	19.5	15.1	(34)
(35)	Mean Daily Temperature Range	MDBR		12.4	11.9	12.0	11.8	11.8	12.2	12.3	11.6	11.7	11.5	11.9	11.9	(35)
(36)		5% DB	MCDBR	15.2	14.9	14.1	13.7	13.3	13.5	13.2	12.5	13.0	12.7	13.6	14.9	(36)
(37)			MCWBR	8.0	7.3	6.0	5.3	5.0	4.9	4.6	4.3	4.5	4.8	6.3	7.7	(37)
(38)		5% WB	MCDWB	13.4	12.7	12.4	11.9	11.8	12.0	12.3	11.6	11.6	11.4	12.6	12.6	(38)
(39)			MCWBR	7.9	7.4	6.6	6.1	5.5	5.2	5.7	5.4	5.3	5.6	6.7	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.267	0.308	0.372	0.399	0.446	0.489	0.566	0.527	0.452	0.395	0.320	0.282	(40)
(41)		taud		2.537	2.403	2.194	2.133	2.012	1.901	1.722	1.828	2.026	2.184	2.428	2.514	(41)
(42)		Ebn at Noon		973	952	905	892	849	809	745	768	814	838	894	933	(42)
(43)		Edh at Noon		83	105	140	155	177	197	234	208	164	129	92	81	(43)
(44)	All-Sky Solar Radiation	RadAvg		3.78	4.72	5.89	6.93	7.50	7.97	7.58	6.88	5.88	4.76	3.87	3.43	(44)
(45)		RadStd		0.23	0.28	0.31	0.32	0.30	0.32	0.21	0.24	0.31	0.24	0.24	0.18	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page